

SIMATS ENGINEERING
Department of Computer Science & Engineering
ASSESSMENT TOOL 2-COMPARATIVE ANALYSIS

Course Code:	CSA12	Course Title:	Computer Architecture
CO Assessed:	CO4	Assessment:	ASSESSMENT TOOL2- COMPARATIVE ANALYSIS
Weightage:	30%	Total Marks:	25
CO4 Statement:	Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics.		
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Department:	CS & BIOSCIENCE	Date:	

ANNEXURE A
COMPARATIVE ANALYSIS

1. Compare L1, L2, and L3 cache levels in terms of latency, bandwidth, and throughput, and analyze their impact on processor performance.
2. Differentiate SRAM and DRAM by evaluating their latency, bandwidth, and throughput characteristics in cache and main memory design.
3. Compare virtual memory with paging and cache memory mechanisms in terms of access latency and system throughput.
4. Analyze the differences between NAND Flash and DRAM with respect to latency, bandwidth, and suitability for hierarchical memory systems.
5. Compare MRAM and RRAM technologies by examining their performance in terms of access time, throughput, and energy efficiency.

Code of Conduct:

I D. Srujana Bai (192571067) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

D. Srujana Bai
Signature of the student:

MARKS ALLOCATION

	Scope of Items (20%)	Comparison Depth (25%)	Use of Evidence (20%)	Critical Thinking (20%)	Clarity of Presentation (15%)	
Q1	1	1.25	0.5	1	0.75	4.5
Q2	0.5	1	0.5	1	0.75	3.75
Q3	0.5	1	1	1	0.5	4
Q4	1	1.25	1	0.5	0.5	4.25
Q5	1	1	1	0.5	0.75	4.25

20.75

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Q1. Compare L1, L2, and L3 Cache Levels in Terms of Latency, Bandwidth, and Throughput, and Analyze Their Impact on Processor Performance

Introduction

Cache memory is a small, high-speed memory located close to the CPU. Modern processors use a hierarchy of caches (L1, L2, and L3) to reduce the time required to access frequently used data.

Comparative Analysis

Parameter	L1 Cache	L2 Cache	L3 Cache
Location	Closest to CPU core	Between L1 and L3	Shared among cores
Size	32 KB – 128 KB	256 KB – 2 MB	4 MB – 64 MB
Latency	1–4 cycles	4–12 cycles	20–50 cycles
Bandwidth	Very High	High	Moderate
Throughput	Highest	High	Moderate
Access Speed	Fastest	Faster	Slower than L1 & L2
Cost per Bit	Highest	Medium	Lower

Evidence and Analysis

- L1 cache stores the most frequently accessed instructions and data.
- L2 cache acts as a backup when data is unavailable in L1.
- L3 cache reduces costly accesses to main memory.

Example

Suppose a processor needs data:

- L1 hit → ~1 ns
- L2 hit → ~3 ns
- L3 hit → ~10 ns
- DRAM access → ~100 ns

This shows how cache hierarchy significantly reduces access delay.

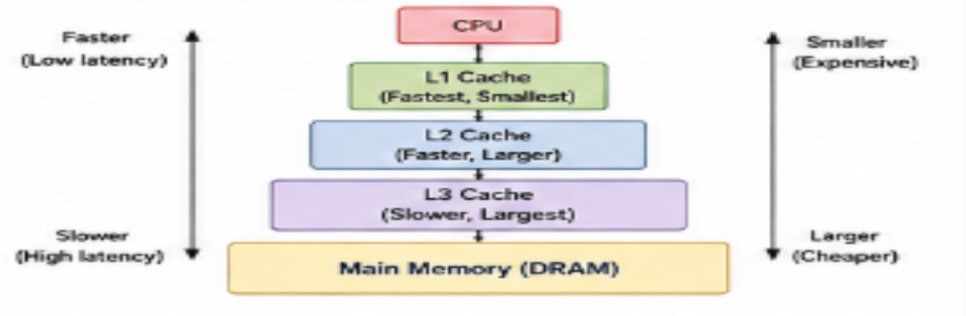
Critical Thinking

A larger cache reduces miss rate but increases cost and power consumption. Designers therefore balance cache size and speed. Modern multi-core processors rely heavily on shared L3 cache to improve inter-core communication.

Conclusion

L1 provides the lowest latency and highest throughput, L2 offers a balance between size and speed, and L3 improves overall system efficiency by minimizing expensive memory accesses. Together they significantly enhance processor performance.

1. L1, L2, AND L3 CACHE LEVELS – COMPARISON			
Parameter	L1 Cache	L2 Cache	L3 Cache
 Size	32KB – 128KB	256KB – 2MB	4MB – 64MB
 Latency	1 – 4 cycles	4 – 12 cycles	20 – 50 cycles
 Bandwidth	Very High	High	Moderate
 Throughput	Highest	High	Moderate
 Location	Closest to CPU core	Between L1 and L3	Shared among cores
 Cost per bit	Highest	Medium	Lower
 Impact on Performance	Reduces CPU wait time significantly	Acts as backup to L1	Minimizes costly memory accesses



Q2. Differentiate SRAM and DRAM by Evaluating Their Latency, Bandwidth, and Throughput Characteristics in Cache and Main Memory Design

Introduction

SRAM (Static RAM) and DRAM (Dynamic RAM) are two major memory technologies used in computer systems. SRAM is mainly used in cache memory, while DRAM is used as main memory.

Comparative Analysis

Parameter	SRAM	DRAM
Full Form	Static RAM	Dynamic RAM
Storage Method	Flip-flops	Capacitors
Refresh Required	No	Yes
Latency	Very Low	Higher
Bandwidth	High	Moderate
Throughput	High	Moderate
Cost	Expensive	Cheap
Density	Lower	Higher

Parameter	SRAM	DRAM
Power Consumption	Higher	Lower
Usage	Cache Memory	Main Memory

Evidence and Analysis

SRAM

- Access time: 1–5 ns
- Used in L1, L2, and L3 caches
- Provides fast data delivery

DRAM

- Access time: 50–100 ns
- Used as system RAM
- Supports larger storage capacities

Critical Thinking

Using SRAM as main memory would provide excellent speed but would be extremely expensive and consume significant chip area. DRAM offers a practical balance between capacity and cost, making it ideal for main memory.

Real-World Example

A computer with:

- 32 MB SRAM cache
- 16 GB DRAM main memory

The SRAM handles frequently used data, while DRAM stores larger datasets.

Conclusion

SRAM provides lower latency and higher throughput but is expensive. DRAM provides large storage capacity at lower cost, making both technologies essential in hierarchical memory design.

2. SRAM vs DRAM – COMPARISON		
Parameter	SRAM	DRAM
Full Form	Static RAM	Dynamic RAM
Storage Cell	Flip-Flop	Capacitor
Refresh	Not Required	Required
Latency	1 – 5 ns (very low)	50 – 100 ns (higher)
Bandwidth	High	Moderate
Throughput	High	Moderate
Power Consumption	Higher	Lower
Density	Lower	Higher
Cost	Expensive	Cheaper
Usage	Cache Memory (L1, L2, L3)	Main Memory (System RAM)

SRAM Cell (6 Transistors)	DRAM Cell (1 Transistor + Capacitor)
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SRAM → Faster but expensive (used in cache)
DRAM → Slower but cheaper (used in main memory)

Q3. Compare Virtual Memory with Paging and Cache Memory Mechanisms in Terms of Access Latency and System Throughput

Introduction

Virtual memory and cache memory improve system performance but serve different purposes. Cache memory accelerates CPU access, while virtual memory expands usable memory through disk storage.

Comparative Analysis

Parameter	Cache Memory	Virtual Memory (Paging)
Purpose	Increase speed	Increase memory capacity
Storage Medium	SRAM	Disk/SSD
Latency	Very Low	High
Throughput	Very High	Lower
Managed By	Hardware	Operating System
Capacity	Small	Very Large
Cost	High	Low

Evidence and Analysis

Cache Memory

- Access time: 1–20 ns
- Reduces CPU waiting time

Virtual Memory

- SSD access: 50–100 μ s
- HDD access: several milliseconds

Critical Thinking

While cache memory directly improves processor performance, virtual memory improves multitasking by allowing applications larger than physical RAM to execute. However, excessive paging causes thrashing, reducing system throughput.

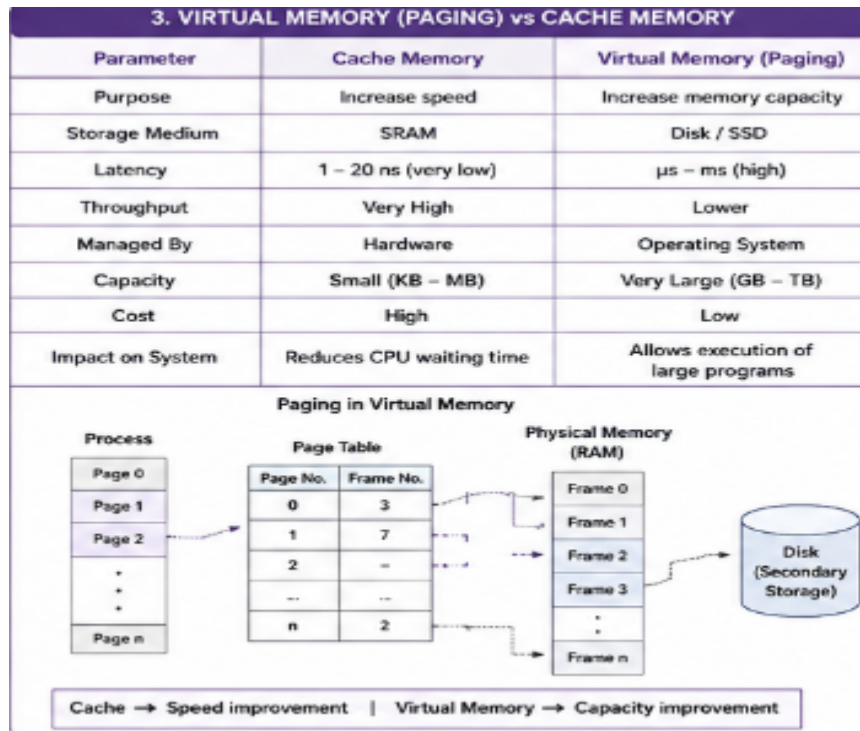
Example

When RAM becomes full:

1. OS moves inactive pages to disk.
2. Required pages are brought back when needed.
3. This process introduces significant latency.

Conclusion

Cache memory focuses on speed and throughput, whereas virtual memory focuses on capacity and efficient memory utilization. Both mechanisms complement each other in modern computer systems.



Q4. Analyze the Differences Between NAND Flash and DRAM with Respect to Latency, Bandwidth, and Suitability for Hierarchical Memory Systems

Introduction

NAND Flash and DRAM are widely used storage technologies. DRAM serves as volatile main memory, whereas NAND Flash is used in SSDs and persistent storage.

Comparative Analysis

Parameter	NAND Flash	DRAM
Volatility	Non-volatile	Volatile
Latency	High	Low
Bandwidth	Moderate	High
Throughput	Moderate	High
Power Consumption	Low	Higher
Storage Density	Very High	Moderate
Cost per Bit	Low	Higher
Usage	SSDs, Storage	Main Memory

Evidence and Analysis

NAND Flash

- Read latency: 25–100 μ s
- Write latency: 200–1000 μ s
- Retains data without power

DRAM

- Access latency: 50–100 ns
- Extremely fast read/write operations

Critical Thinking

DRAM cannot replace NAND Flash because it loses data when power is removed. NAND Flash cannot replace DRAM because its latency is much higher. Therefore, both technologies occupy different levels of the memory hierarchy.

Memory Hierarchy Placement

CPU → Cache → DRAM → NAND Flash SSD → HDD

Conclusion

DRAM delivers superior latency, bandwidth, and throughput, while NAND Flash offers persistent storage and high density. Together they create an efficient hierarchical memory system.

4. NAND FLASH vs DRAM – COMPARISON		
Parameter	NAND Flash	DRAM
Volatility	Non-volatile	Volatile
Latency	Read: 25–100 μ s Write: 200–1000 μ s	50 – 100 ns
Bandwidth	Moderate (100 – 400 MB/s)	High (10 – 50 GB/s)
Throughput	Moderate	High
Power Consumption	Low	Higher
Density	Very High	Moderate
Cost per bit	Low	Higher
Suitability	SSD, Storage Devices	Main Memory (RAM)

Memory Hierarchy Position

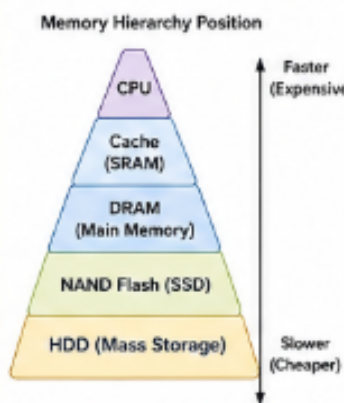


Diagram illustrating the Memory Hierarchy Position:

- CPU (Fastest, Most Expensive)
- Cache (SRAM)
- DRAM (Main Memory)
- NAND Flash (SSD)
- HDD (Mass Storage) (Slowest, Cheapest)

Speed and Cost Trend: Faster (Expensive) at the top, Slower (Cheaper) at the bottom.

DRAM → Fast and volatile (main memory) | NAND Flash → Non-volatile and dense (storage)

Q5. Compare MRAM and RRAM Technologies by Examining Their Performance in Terms of Access Time, Throughput, and Energy Efficiency

Introduction

MRAM (Magnetoresistive RAM) and RRAM (Resistive RAM) are emerging non-volatile memory technologies designed to overcome the limitations of traditional memory systems.

Comparative Analysis

Parameter	MRAM	RRAM
Storage Mechanism	Magnetic states	Resistance changes
Volatility	Non-volatile	Non-volatile
Access Time	Very Fast	Fast
Throughput	High	Moderate to High
Energy Efficiency	High	Very High
Endurance	Excellent	Good
Density	Moderate	Very High
Scalability	Good	Excellent

Evidence and Analysis

MRAM

- Access time: 10–30 ns
- High endurance ($>10^{15}$ write cycles)
- Suitable for high-speed embedded systems

RRAM

- Access time: 10–100 ns
- Low power consumption
- Suitable for AI accelerators and IoT devices

Critical Thinking

MRAM is ideal where speed and durability are critical. RRAM is preferred where low power consumption and high storage density are more important. Future computing systems may integrate both technologies depending on application requirements.

Real-World Applications

- MRAM: Automotive systems, aerospace electronics, industrial controllers.
- RRAM: AI hardware, IoT devices, neuromorphic computing.

Conclusion

MRAM offers superior throughput and endurance, while RRAM provides excellent energy efficiency and scalability. Both technologies have strong potential to become key components of next-generation memory architectures.

Overall Conclusion

The memory hierarchy—from L1/L2/L3 caches, SRAM, DRAM, virtual memory, NAND Flash, MRAM, and RRAM—is designed to balance latency, bandwidth, throughput, capacity, cost, and

energy efficiency. Each technology occupies a unique position in the hierarchy, ensuring optimal system performance while maintaining cost-effectiveness and scalability.

Parameter	MRAM	RRAM
Technology	Magnetoresistive RAM	Resistive RAM
Storage Mechanism	Magnetic states	Resistance changes
Access Time	10 – 30 ns	10 – 100 ns
Throughput	High	Moderate to High
Energy Efficiency	High	Very High
Endurance	Excellent ($>10^{15}$ cycles)	Good ($10^8 - 10^{12}$ cycles)
Density	Moderate	Very High
Best Suited For	Embedded systems, Automotive, Aerospace	AI hardware, IoT, Neuromorphic computing

Key Takeaway

 **MRAM** → Faster speed & high endurance

 **RRAM** → Better energy efficiency & scalability

 Both are promising non-volatile memories for future systems

Common Properties

- ✓ Non-volatile
- ✓ High speed
- ✓ Low power
- ✓ Future memory technologies